Species classification using transformer models in a multimodal setting.

Technical Report for 4343MMEBX

Group 1

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Abstract

Biodiversity monitoring plays a pivotal role in defending against climate change and invasive species (Gharaee et al., 2025a). Gharaee et al. (2025a) introduced the BIOSCAN-5M dataset, where the multimodal structure of the dataset enables further innovation in machine learning, and demonstrated the effectiveness of pre-trained embeddings for taxonomic classification in multimodal setting. Where using transformer models to combine both DNA and image embeddings using pre-trained weights from larger datasets could be an effective way in improving species classification (Gharaee et al., 2025a). A study is conducted that combines the visual and genetic information by using the pre-trained DINOv2 and DNABERT6 models as encoders, where the embeddings are fused with a simple concatenation to train a multimodal species classification model. We compare the fused embeddings with the baseline embeddings and compare a hierarchical and single-label approach.

The results obtained indicate that simple concatenation without any auxiliary objectives does not significantly improve the results over the embeddings created by DNABERT6 alone. Where for the validation dataset the accuracy of DNABERT6 is 0.86, compared to 0.56 for the fusion model, but the macro F1 score for DNABERT6 is 0.54 compared to 0.56 for the fusion model. These metrics are from the hierarchical approach, and show that hierarchical learning performs better than single-label. In addition, a t-SNE visualisation clearly shows that hierarchical training gives more defined clusters for the taxonomic structure compared to single-label. Even though the fusion model does not perform much better than DNABERT6, these experiments show that transformer models perform well in encoding image and DNA features through fine-tuning, to create a relatively high performance model that is able to perform multi-class species classification in a multimodal setting.

1 INTRODUCTION

Biodiversity plays an important role in sustaining ecosystems and provides a natural defence against disturbances such as climate change and invasive species (Gharaee et al., 2025a). As part of an ongoing worldwide effort to monitor the biodiversity of insects, Gharaee et al. (2023) introduced the BIOSCAN-1M insect dataset, which pairs DNA with images to develop machine learning tools for automatic classification. However, this dataset contains family taxonomic labels, and does not fully utilize the multimodal nature of the dataset. Gharaee et al. (2025a) present the BIOSCAN-5M dataset, which expands the dataset by including more complete taxonomy and genetic information over 5 million arthropod specimens. Compared to BIOSCAN-1M, this dataset provides more high-resolution images and DNA barcodes, along with taxonomic ranks, species size, and other metadata. The multimodal structure of BIOSCAN-5M further innovates machine learning. Gharaee et al. (2025a) conduct experiments using this multi-modal information instead of only the image modality used in Gharaee et al. (2023), where they train BarcodeBERT (Arias et al., 2024) on the DNA barcode data. Where a zero-shot transfer learning task is performed using zero-shot clustering representation embeddings. This approach demonstrated the effectiveness of pre-trained embeddings in clustering data, even in the absence of ground-truth. As a result, a shared embedding space is learned across three modalities in the dataset, RGB images, taxonomic labels and DNA barcodes, for fine-grained taxonomic classification.

The research presented in this report focuses on creating a multimodal pipeline using the image and DNA barcodes from the BIOSCAN-5M dataset to perform species classification. We develop two unimodal baselines and a multimodal model that combines DINOv2-small and DNABERT6. DINOv2-small was selected since it achieved high performance on species classification tasks, as stated by Gharaee et al. (2025a), while DNABERT also showed a similarly strong performance. The image and DNA models are used to perform species classification and evaluated, after which a multimodal version of this model is created by fusing the embeddings from the encoders through concatenation. This research investigates the performance of transformer-based models in a multimodal setting for insect species classification, and evaluates whether a combined fine-tuning of image and DNA modalities improves embedding quality compared to independently trained unimodal baselines, as measured by top-1 accuracy and F1-score.

2 RELATED WORK

Prior works have developed machine learning models for taxonomic classification of images and DNA, where Gharaee et al. (2025a) use CLIBD (Gong et al., 2025), to introduce a multimodal approach that combines both modalities, using CLIP-style contrastive learning to align images, DNA barcodes and text-based representations of the taxonomic labels in a unified embeddings space. This type of method allows for the classification of both known and unknown insect species, without task-specific fine-tuning. CLIP-style learning is a current state-of-the-art approach for unseen fine-grained species classification. Gharaee et al. (2025a) created BIOSCAN-5M. A large, high-quality biological training and validation dataset for multimodal models, demonstrating strong performance in aligning image and DNA representations for classification tasks, which piqued our interest for the use of BIOSCAN-5M in this project.

Ji et al. (2021) proposed a BERT-based model for extracting semantic features from DNA input sequences, which has become a widely used foundation in biological machine learning tasks. Zhou et al. (2024) proposed their newest generation of encoders for DNA, DNA-BERT-2, which improves its predecessor by using Byte Pair Encoding (BPE) instead of k-mer tokenization, increasing computational efficiency. We intentionally selected DNABERT6, a version of the original DNABERT, as our DNA encoder due to its fundamental features and comparability. This provides a consistent baseline to verify that improvements are due to multimodal applications rather than recent developments in model design and efficiency. While newer models such as DNABERT-2 may result in stronger representations, incorporating them would introduce additional variables beyond the focus of this study. Oquab et al. (2024a) showed that visual image transformer models (ViTs), if trained on enough data from varied sources, could provide general visual embeddings useful for a variety of general-purpose tasks. Due to its wide generalization and performance as stated by Gharaee et al. (2025a), it was selected as our visual encoder to be able to easily fine-tune it for our needs.

3 DATA

Table 1. Overview of the BIOSCAN-5M dataset splits.

Dataset name	Modality	Images	Barcodes	Usage Purpose
cropped_256_train	Image + DNA + Labels	289,203	118,051	Training
cropped_256_eval	Image + DNA + Labels	14,757	6,588	Validation

The BIOSCAN-5M dataset consist of insect specimen images and their corresponding metadata, such as taxonomic, geographic and DNA barcodes (Gharaee et al., 2025b). For the project the training and validation dataset are used, which are ”cropped_256_train” and ”cropped_256_eval” obtained from Hugging Face. These datasets contain cropped images that were cropped using the BIOSCAN-5M cropping tool (Gharaee et al., 2025b). An overview of these datasets are seen in table 1, that shows that the number of images and DNA barcodes differ, with there being about half the number of barcodes compared to images. These samples cover around 11,846 species in total (Gharaee et al., 2025a).

3.1 Pre-processing

The data was preprocessed using the DINOv2 AutoImageProcessor which prepares the images to make sure they can be used as input for the model by applying operations such as centering, normalization and cropping (huggingface.co, 2026a). The DNA barcode is split into a string of 6-mers, using a sliding window from start to end, separated by spaces. This string is then used as input for the DNABERT6 AutoTokenizer. The AutoTokenizer then converts the 6-mers into tokens for the transformer model, applying any padding or truncation to make sure the input tensor contains 512 tokens (huggingface.co, 2026b).

3.2 Train/Validation/Test Split

The previously mentioned data splits (Tab. 1) were loaded using the ”dataset.py” script, which directly downloads the .zip files from Hugging Face and loads them using python. The validation set contains the new samples of the same species present in the training data, to prevent a biased validation on already seen samples (Gharaee et al., 2025b). A predetermined set seed was used to iterate over the datasets, to make the dataset loading and runs deterministic and consistent. Since these splits are precomputed, only the loading is affected by this seed.

4 METHODOLOGY

The task is to perform species classification using transformer-based encoders, specifically for creating visual and DNA encodings of barcodes and an image of the respective organism, for multi-class classification. The model serves as a proof of concept for evaluating transformer performance in a multimodal setting for this problem space. There are two input modalities in the dataset, images and DNA, with multiple textual labels defining the class, family, order, genus and species of a given image and barcode of insect specimens. The images consist of RGB lab images of insects on glass slides, with DNA barcodes for the respective specimen of 400-1300 nucleotides in length. For the multimodal setting, we created concatenated image and DNA embeddings from their respective encoders to classify the species labels using a classification network. The output labels are the species scientific name, e.g. *Sus scrofa*, which are converted into class indices for the model to predict. The first class index starts at zero and counts up from there for each new string it receives. This is done in the first part of the model initialization, which are then stored in a file for the model to be able to map its classifications back to the original labels it was trained on.

The model has two training settings, ”single-label” and ”hierarchical”. The single-label setting trains the model only on the taxonomic species labels, doing no layer freezing and simply just fine-tuning based on classification of class indices. The hierarchical training setting trains the model on five groups of taxonomic labels, and freezing certain model weights to retain information of higher-level taxonomic groups. It starts off at class level and ends at species level. Where, after switching between label types, the final layer in the classification head is replaced with a new layer that matches the number of output classes for that label.

4.1 Model Architecture

For the encoders for the image and DNA from the dataset, we decided to use the relatively new and state-of-the-art DNABERT6 and DINOv2, which are both transformer architectures (Zhou et al., 2024; Oquab et al., 2024a). Transformer models were selected because they excel at transfer between problem spaces, meaning it is easy to fine-tune them to our problem space and still have a high model performance. Where DINOv2 was stated by Gharaee et al. (2025a) to be the best performing vision encoder for the Zero-shot transfer-learning experiments, and DNABERT transformer models had a similar strong performance. For the goal of species classification, these models are initialized using their pre-trained weights and then further fine-tuned separately and together in a unimodal and multimodal setting to evaluate their use on taxonomic species classification.

The classification head is a simple neural network that maps the embeddings to the label indices the model is actively training on. It consists of 2 layers of hidden size 256 and 128 in sequence, the first and second layer are followed by a 20% dropout after a layer normalization and ReLU activation. The final layer output size is relative to the number of output classes, which only changes in the hierarchical setting. Table 2 shows a more detailed description. An overview of the complete multimodal architecture, including encoders, concatenation, and classification, can be seen in Figure 1.

Table 2. Classification head layer architecture.

Layer	n_input	n_output	Normalization	Activation	Dropout
Input	embedding_size	256	Yes	ReLU	20%
Hidden	256	128	Yes	ReLU	20%
Output	128	n_classes	No	Softmax	None

There are two settings in our model apart from hierarchical training, named ”ds_rand” and ”augment” in the training arguments. ”ds_rand” stands for dataset randomization, and while this is enabled during a run, the dataset is shuffled before each epoch to increase the overall coverage on the dataset. Without this parameter enabled, the model will simply sequentially iterate over the dataset and starts at the beginning again at the start of each new epoch. The parameter ”augment” stands for augmentation, where a different augmentation operation is done for each modality. When a model is run using this option for the DNA augmentation, before creating the k-mers, a random 510 length part of the barcode sequence is taken if the sequence is longer 510 nucleotides. This ensures the truncated sequence is more representative of the original barcode, and still fits in the 512 token max of BERT models, accounting for the [CLS] and [SEP] tokens. For the image augmentation, three operations are independently applied with 0.5 probability of applying; horizontal flipping, vertical flipping, and applying Gaussian noising to the image. This is to make the images more generalized overall. Without augmentation, the DNA sequences are only truncated at the end, and the images are fed into the AutoImageProcessor without any additions.

4.2 Alignment and coordination/fusion Strategy

In the fusion model, the output embedding of DINOv2 and DNABERT6 are concatenated together into a single fused embedding. The embedding of DNABERT6 is a tensor of 768 length, and DINOv2 one of 384 length. Combining these two embeddings results in a tensor of length 1152.

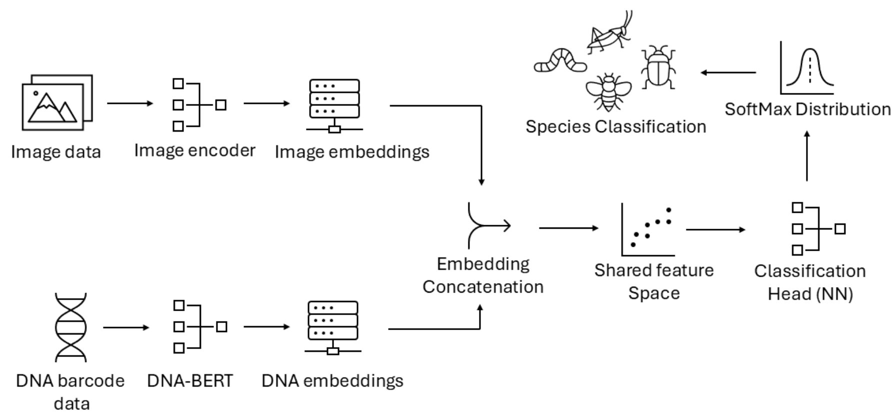


Figure 1. Diagram of the proposed multimodal architecture.

4.3 Implementation Details

The models were implemented using the PyTorch and Hugging Face packages in python (Paszke et al., 2019; Lhoest et al., 2021; Wolf et al., 2020). Each model was run using the same hyperparameters, which can be seen in table 3. The parameters defining layer freezing and epochs per taxonomic label for hierarchical training can be seen in table 4. Batch size is the number of datapoints per step the model trains on, where steps per epoch is the number of batches per epoch. K-mer size is a variable parameter that defines the k-mer size sampled from the DNA barcodes, which retains a value of six for DNABERT6. Frozen layers defines how many layers are frozen in the encoders before the model starts training on that label. A value of 'None' means no freezing takes place for that label. The epochs per taxonomic label are the number of epochs the model trains on that specific label. A random seed is set at the start of each model run, and is the same for all runs to ensure reproducibility, where it is applied to every python package that uses random operations in our implementation. The GitHub repository contains all scripts for generating the graphs and results present in the report. The scripts are fully configured to our hyperparameters, and need some tweaking to be able to run on other parameters.

Table 3. Main hyperparameters

Parameter	Value
Framework	PyTorch
Optimizer	Adam
Learning rate	5e-5
Batch size	16
Steps per epoch	200
Epochs	100
K-mer size	6
Loss function	Cross-Entropy
GPU 1	NVIDIA GeForce RTX 4060
GPU 2	NVIDIA GeForce GTX TITAN X
Random seed	202667

Table 4. Hierarchical training setting hyperparameters.

Taxonomic level	Epochs	Frozen layers
Class	2	None
Order	4	1
Family	10	4
Genus	25	5
Species	100	9

5 EXPERIMENTS

5.1 Experimental Setup

To evaluate the strength of each model, we run the unimodal baselines on single-species (base) and hierarchical (hier) settings. We run both of these training schemes twice, one using augmentation and dataset randomization and one without, and repeat this for both the single-species and hierarchical for unimodal baseline models. We define four training schemes: ”base”, ”hier”, ”base_mod” and ”hier_mod” for the unimodal baselines. Where ”base” stands for single-species training, ”hier” stands for hierarchical training, and ”mod” for the combination of both data augmentation and dataset shuffling. For the multimodal fusion model, both training schemes have the augmentation and dataset shuffling. We evaluate the training progress of each model for each batch, where the loss for the training dataset and the validation dataset is saved, which is averaged for each epoch for plotting the training curve. On the final epoch, the training and validation classification accuracy and macro F1-score is also evaluated over the entire epoch.

To assess the performance of each model, the unimodal baselines are evaluated to determine how the single-species or hierarchical settings perform in a unimodal situation, with augmentation and dataset shuffling alterations compared for these settings. The best performing training scheme is then used to train the multimodal setting, focusing on the validation classification accuracy and F1. Based on the performance of the non-modified unimodal baselines, we decided to train the multimodal model using only the modified setting.

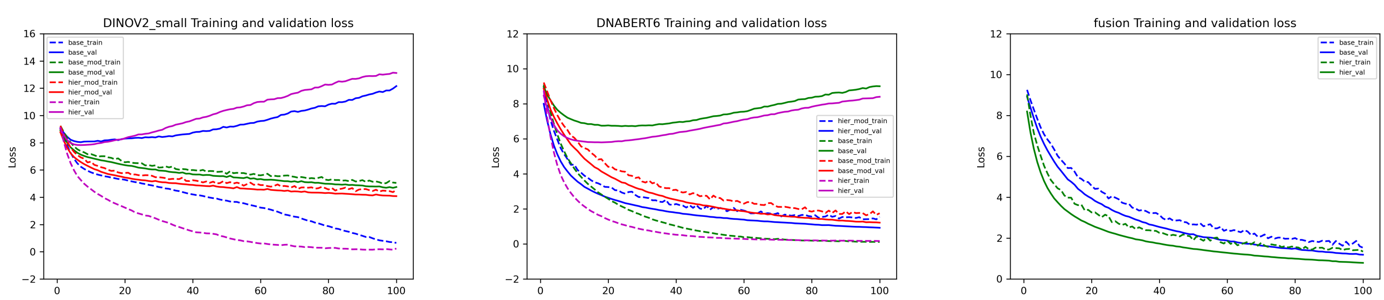
5.2 Evaluation Metrics

Three metrics are calculated for each experimental run. The classification accuracy and macro F1-score are calculated on the final training epoch for the training and validation dataset, where in addition the loss is calculated for the training and validation dataset over each epoch. The training accuracy and validation accuracy were used to compare the accuracy of the trained models on both seen an unseen datapoints of known classes, to validate whether or not the model can generalize. The macro F1-score is used to give a more realistic metric to account for the potential imbalance in the dataset distribution. The loss is recorded to show the general trend in loss for both the training and validation dataset over each epoch, rather than a single evaluation on the final epoch for the accuracy and F1 to visualise performance during training. A fixed subset of the validation dataset was used to calculate the loss and other validation metrics, to be able to compare the validation metrics between model runs and epochs. Additionally a t-SNE visualization is created for the final fusion model runs for 3200 data points, coloured for genus and species in seperate plots.

6 RESULTS

Quantitative Results

A visualization of training and evaluation loss over training epochs can be seen in figure 2. The base and modified runs showed a substantially different training curve, where the non-modified runs all had lower training loss in comparison, the validation loss went up after a while and continued increasing. The modified runs, which had dataset shuffling and data augmentation, had their training and validation loss decrease in a consistent manner. In all loss plots (Fig. 2), we can see that the loss of the hierarchical runs decreased faster than the single-label runs, in both the unimodal and multimodal runs.



(a) (b) (c)

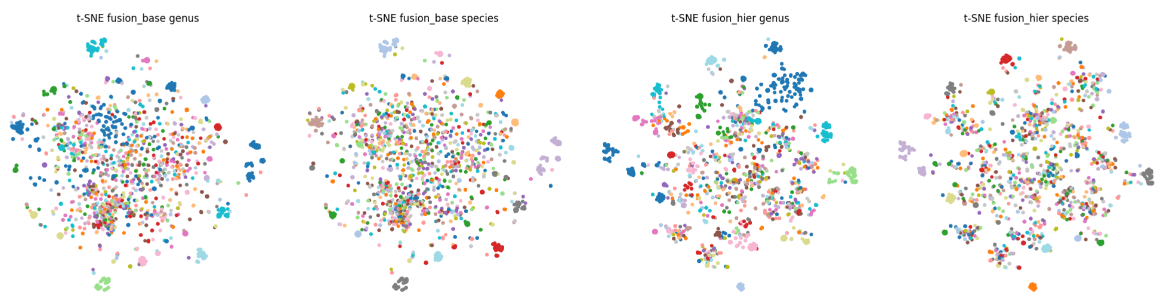
Figure 2. Training curve showing the mean loss per epoch for the training (dotted line) and validation (solid line) dataset, for each experimental run. (a) shows the loss curve for DINOv2-small, for the ”base” and ”hier” setting, with or without modifications. (b) shows the loss curve for DNABERT6, for the ”base” and ”hier” setting, with or without modifications. (c) shows the the loss curve for the multimodal model, for the ”base” and ”hier” setting with modifications.

The metrics in table 5 show the training and validation classification accuracy and macro F1-score for all model runs, with the best accuracy and F1-score highlighted for both the training and validation metrics. DINOv2 achieved the highest training F1-score for hierarchical unmodified, but the validation accuracy and F1-scores were much lower in comparison to its modified counterpart. DNABERT6 had the highest training accuracy for single-label unmodified, but similarly to the DINOv2 runs, the validation accuracy and F1 were much higher for its modified counterpart. The fusion model had a lower training accuracy and F1 score compared to DINOv2 and DNABERT, while the validation F1 score was higher compared to the unimodal model for hierarchical modified, where only the accuracy was slightly lower compared to DNABERT6.

Table 5. Experimental results for the training and validation dataset.

	Train						Validation					
	DINOV2_small		DNABERT6		Fusion		DINOV2_small		DNABERT6		Fusion	
	Accuracy	F1	Accuracy	F1	Accuracy	F1	Accuracy	F1	Accuracy	F1	Accuracy	F1
Single-label	0.94	0.83	**0.97**	0.93	—	—	0.10	0.01	0.52	0.17	—	—
Single-label modified	0.24	0.02	0.80	0.44	0.83	0.84	0.25	0.02	0.83	0.47	0.80	0.52
Hierarchical	0.96	**0.96**	0.96	0.91	—	—	0.10	0.01	0.57	0.21	—	—
Hierarchical modified	0.31	0.05	0.82	0.50	0.85	0.87	0.33	0.05	**0.86**	0.54	0.82	**0.56**

The scatter plots (Fig. 3) show a t-SNE representation of 3200 data points from the validation dataset, showing the embedding space of the single-label and hierarchical fusion models. These plots show that the embeddings properly represented both genus and species clustering, being more well-defined in the hierarchical training runs in (c) and (d), compared to (a) and (b). These plots also show that different species embeddings with same genus cluster together, and that species abundant in the dataset had better defined clusters compared to less abundant species.



(a) **(b)** **(c)** **(d)**

Figure 3. tSNE scatterplots visualising the multimodal embeddings for both the hierarchical and single-label trained fusion model. (a) shows the tSNE with "fusion_base" embeddings coloured by genus labels. (b) shows the tSNE with "fusion_base" embeddings coloured by species labels (c) shows the tSNE with "fusion_hier" embeddings coloured by genus labels. (d) shows the tSNE with "fusion_hier" coloured by species labels.

7 DISCUSSION

The results of this study show that DNABERT6 had the highest validation classification accuracy of 0.86, with the multimodality fusion model having the highest macro F1-score of 0.56 (Tab. 5). Both results were from the hierarchical setting including modifications. Any experiments not run using the modification settings showed significantly lower validation scores compared to the training set, which is likely due to dataset shuffling, as without randomization, the model trains only on a static part of the dataset, drastically decreasing the coverage of the actual dataset during training. The hierarchical training setting showed increased F1 and classification accuracy in both datasets, which had a positive effect on both the modified and non-modified settings (Tab. 5), most likely attributed to the model learning information about higher-level taxonomic groups early on, thus increasing the models' ability in differentiation between species labels, which can also be seen in the t-SNE visualization (Fig. 3). Comparing the performance of the multimodality fusion model to the unimodal baselines showed that the DNA modality had much higher scores for both the training and validation dataset in comparison to the image modality (Tab. 5). Additionally, the validation accuracy of DNABERT6 is slightly higher than the fusion model for the hierarchical modified run, indicating that the model does not extract much additional information from the method in which the image and DNA modality are fused. Since the DNA embedding is much larger from DNABERT6 (764) compared to DINOv2-small (384), and by the fact DNA is incredibly informative to species identification, this suggests that the visual modality contributes less than the DNA modality in the current fusion setup (Oquab et al., 2024b).

The main limitation of the research is the limited batch size of 16, as the available computing resources limited the amount of data that could be loaded at once. Increasing the batch size would help the model cover more of the dataset to potentially improve the models' performance. Another limitation is that the dataset consist only of only seen species, so the created model architecture is unable to deal with unseen species for classification, specifically because the model predicts class indices of species it was trained on. Changing the manner in which it classifies species by validating the model on an additional unseen set of species could be an improvement in making the model more effective in species classification tasks. Another limitation of this work is the use of simple concatenation for multimodal fusion, which does not explicitly enforce cross-modal alignment. The t-SNE visualization (Fig. 3) shows a class imbalance in the dataset, where species with higher abundance in the dataset give well defined clusters as seen in figure 3 (b) and (d), while in comparison (a) and (c) show that the less abundant species that fall under the same genus do not give well defined clusters, clustering further apart than the higher abundant species.

The ecological relevance of the results, show that adding DNA to species classification models is very effective for improving results, and thus a model could be created and trained on DNA, while not require it during inference. This would increase the overall performance by a large margin, as has also been shown by other studies such as Gong et al. (2025).

7.1 Future works

While our proposed model performs quite well with a relatively low batch size, it would be interesting to see what effect a larger batch size would have on training efficiency and on the classification results. Since DNA barcoding is time- and resource intensive, our proposed method might not be universally applicable to multi-modal classification problems between the visual and DNA modalities. Future work might consist of applying a cross-modal approach, where the image encoder is trained to align with or predict DNA embeddings from the DNA encoder instead of simple concatenation of the resulting embeddings like we have done here. Evidently from the positive impact of hierarchical training settings, future work could look into a way to use the hierarchical structure as an auxiliary objective during training/fine-tuning, which would help the model reinforce the higher-level taxonomic information more. Additionally, using more recent models such as DNABERT-2 as the DNA encoder could result in more information rich embeddings.

8 REPRODUCIBILITY STATEMENT

The full source code can be found at the project's https://github.com/MMblokl/DNAVMM. To reproduce the results in this paper, clone the git repository and download the required packages with uv: "uv sync". Then either use "uv run python" as your run command, or activate the virtual environment "source .venv/bin/activate" and use "python3". If you wish to add a Hugging Face API key or run the model on a server with a storage quota, fill in the "env_example" file. This is optional as the model will work without these settings. After filling in the values, rename "env_example" to ".env".

To run the models, run one of the scripts referring to a specific model: "unimodal_image.py" (DINOv2-small only), "unimodal_dna.py" (DNABERT6 only) or "fusion.py" (multimodal fusion model). Where the run command has the following structure: "python3 script.py run_name/ ds_rand hierarchical augment", where "run_name" refers to the directory where the results will be stored, and "ds_rand", "augment" and "hierarchical" refer to the dataset randomization, data augmentation and hierarchical training scheme respectively. The model scripts automatically load the dataset through the dataset.py script and preprocess their respective modality input. In addition the model scripts contain a set seed to make all results deterministic.

To create the training/validation loss and t-SNE plots, run the "plotcreator.py" and "run_tsne.py" scripts. To create the t-SNE visualizations of a model run, simply run "python3 run_tsne.py run_name", with "run_name" being the same directory created for the model output. To create the loss visualization plots, create a new directory "dir_name" and place the "model_metrics.npy" file from any number of model runs inside. Make sure to rename these metric files to something that identifies what run it came from such as "base" or "base_mod". To create the plots run "python3 plotcreator.py dir_name"

For a more detailed explanation, refer to the README.md in the git repository. This README goes into more detail than this section and include alternative ways of installing dependencies.

9 CONCLUSION

Through these experiments, we find that transformer models perform well in encoding image and DNA features through fine-tuning, enough to create a relatively high-performance model using limited computing resources, that is able to perform multi-class species classification in a multimodal setting.

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